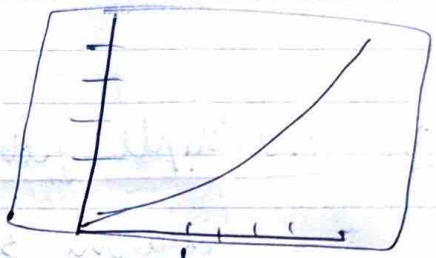
Data Visualisation with Matplotlib

→ ! pip install matplotlib

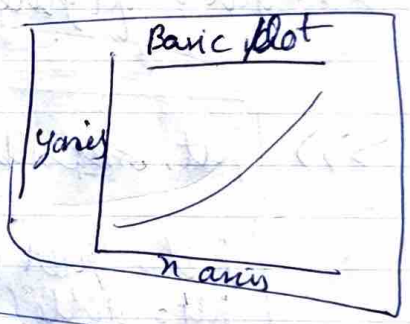
or add it to req.txt.

1 Basic

```
>>> x = [1, 2, 3, 4, 5]
>>> y = [1, 4, 9, 16, 25]
>>> plt.plot(x, y)
```



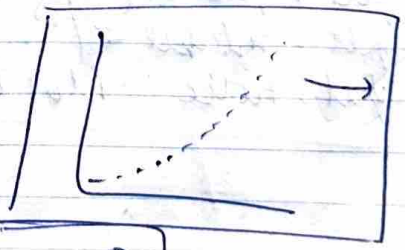
```
>>> plt.xlabel('x axis')
plt.ylabel('y axis')
plt.title('Basic plot')
plt.show()
```



2 Customisation

```
>>> plt.plot(x, y, color='red', linestyle='--')
```

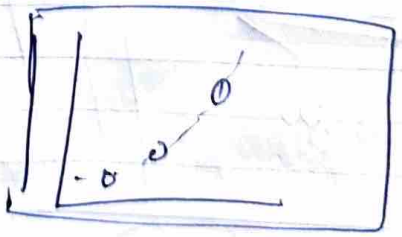
graph



→ in red colour

```
>>> plt.plot(x, y, marker='o')
```

⇒ places a marker at every coordinate



- more parameters

→ linewidth = 3, 4, ...

→ markersize = 2, 3 ...

→ plt.grid(True) → builds a grid behind the existing plot.

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③ Multiple plots

imagine 3 lists x, y, z

```
>>> plt.figure(figsize=(9,5))
```

→ control the length/breadth of plots

parent figure size

```
>>> plt.subplot(1,2,1)
    plt.plot(x, y, color='green')
    plt.title('Plot 1')
```

position

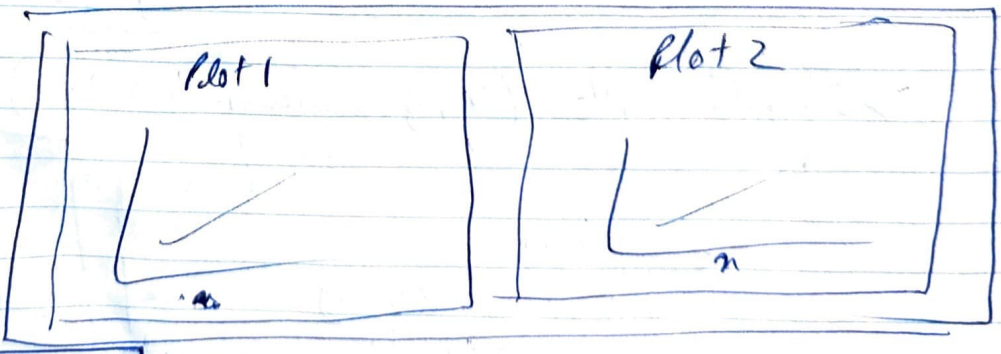
no of rows

no of columns

```
>>> plt.subplot(1,2,2)
    plt.plot(y, z, color='red')
    plt.xlabel('x')
    plt.title('Plot 2')
```

subplot can be (2,2,1) or anything

→ now



(*) The subplot(args) arguments divide the figsize = (,) into respective ratios

2 columns

#

(4) Barplot

→ for something related to categories

```
>>> categories = ['A','B','C']
>>> values = [5, 7, 3]
>>> plt.bar(categories, values, color='red')
```

bar graph

→ similar parameters, plt.title()
plt.xlabel()
plt.ylabel.

---

(5) Histogram

```
>>> data = [1,2,2,3,3,3 . . . . 5.]
>>> plt.hist(data, bins = 5)
```

divide data into 5 buckets.

→ parameters → , color = ' '
, edgecolor = ' '

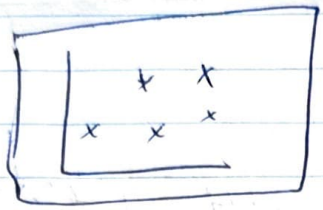
PTO

(6) Scatter plot (#)

x, y . . . -

```
plt.scatter(x, y, color='blue', marker='x')
```

similar parameters, plt.xlabel(), title(), etc



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(7) Pie-chart

labels = [A, B, C]
sizes = [10, 30, 60]
colors = ['red', 'green', 'blue']
explode = (0.2, 0, 0) → moves out the first slice

create

length = len(labels)

```
plt.pie(sizes, labels=labels, colors=colors,
        autopct='%1.1f%%')
```

plt.pie()

shows the %

```
plt.pie(explode=explode, labels= . . . -
        . . . . . )
```

explode → shifts the slices out.
→ Should always be equal to size.

Plotting from data

eg. df-grouped = df.groupby ('Pro')['Val'].sum()

```
df-grouped . plot (kind = 'bar', color = 'teal')
```

grouped data — the type of graph — color of graph

- .reset_index() → resets the index & gives a df.

---

eg ~~df-gr = df.group~~

```
n-df = df.groupby('Pro')['Val'].sum().reset_index()
```

resets index.

	Pro	Val
0	—	—
1	—	—
2	—	—

n-df.

Now, to plot any graph,

```
plt.bar/scatter (n-df['Pro'], n-df['Val'])
```

2 lists (x, y)